ADF Project

Mahender Kore

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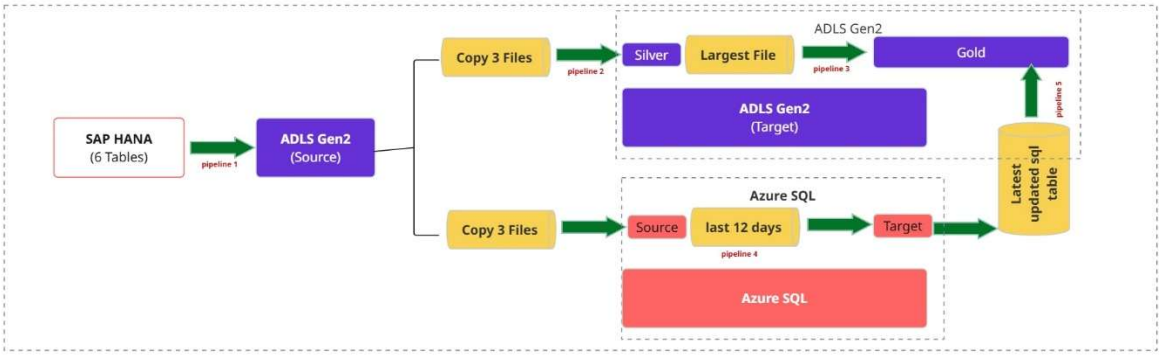
1. Architecture Diagram

Aim of this project is to build an end-to-end data engineering pipeline using Azure Data Factory (ADF) that extracts data from **SAP HANA**, processes it across different layers, and stores it in **Azure Data Lake Storage (ADLS) and SQL Server**.

The architecture follows a **Medallion Architecture (Source → Silver → Gold)** ensuring scalability, performance, and data quality.



Architecture diagram

PART 1 – SAP HANA to ADLS

- Connect SAP HANA to ADF
- Copy 6 tables to Source ADLS

PART 2 – ADLS Processing

- Copy 3 files (MARA, VBRK, VBAP) from Source ADLS to Target ADLS – Silver Layer
- Dynamically identify the largest file
- Copy the largest file into Gold Layer (ADLS)

PART 2 – SQL + Incremental Load

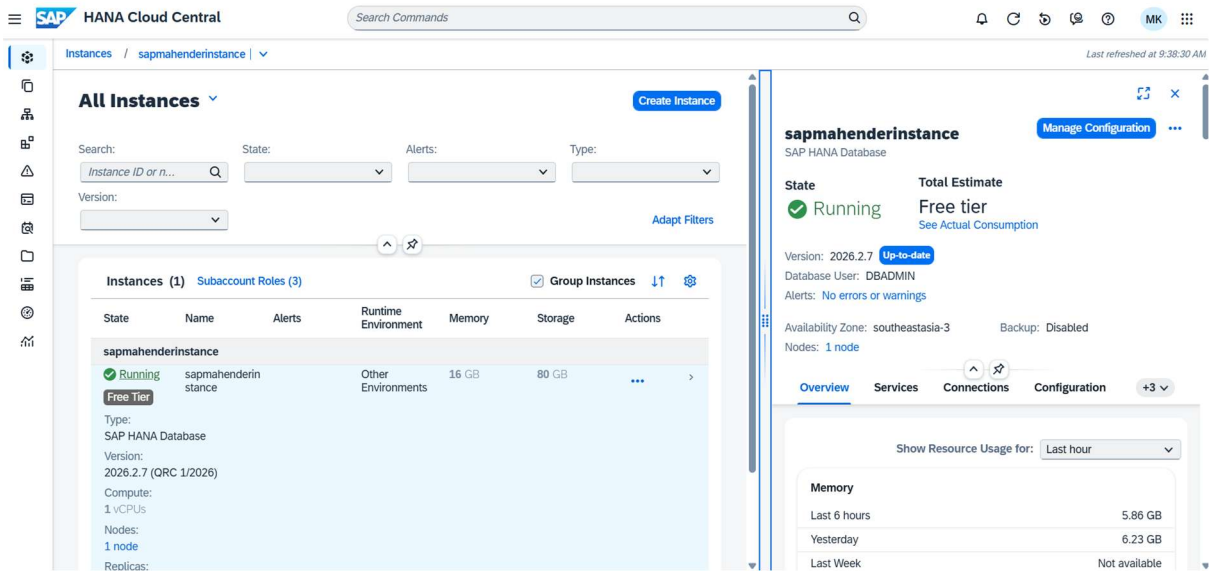
- Copy 3 files (COSTLIST, INVOICE_LINE_ITEM, PRODUCING_STAT_RATIO_DETAIL) from Source ADLS → Source SQL
- Load only last 12 days data into Target SQL – Silver Schema
- Identify latest modified records (MAX LastUpdated)
- Load latest records into Gold Layer (ADLS)

2. SAP HANA Data Creation

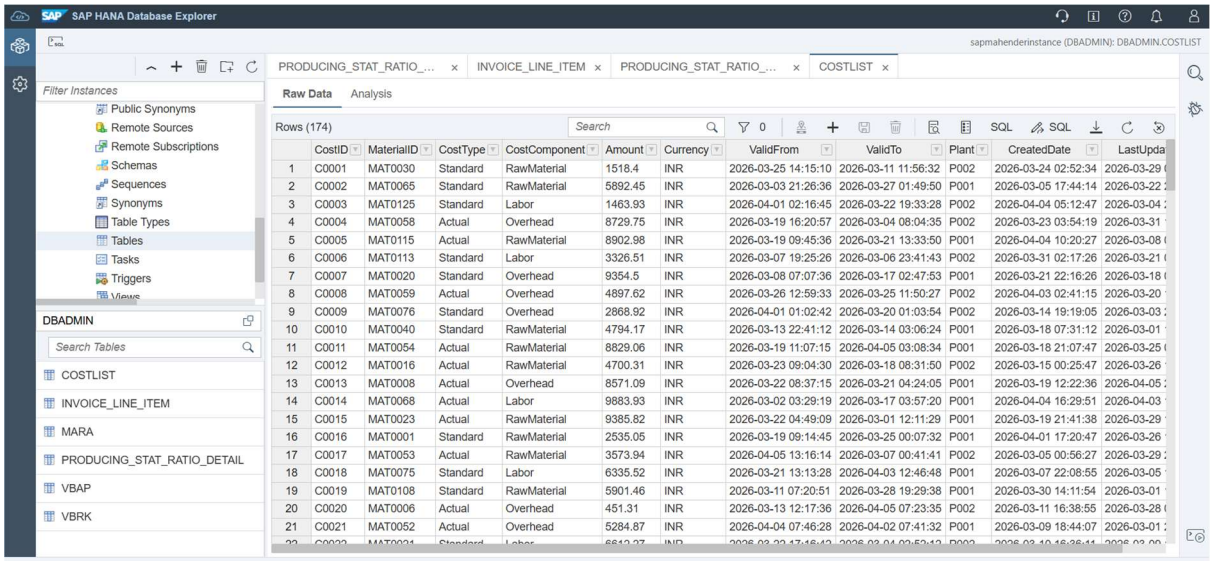
2.1 SAP HANA Table Creation

- Create SAP HANA Trial Account using SAP BTP Cockpit
- Enable SAP HANA Cloud Database
- Open Database Explorer
- Use Import Data Wizard to load data
- Tables Created: MARA, VBAP, COSTLIST, INVOICE_LINE_ITEM, PRODUCING_STAT_RATIO_DETAIL, VBRK

All 6 tables are successfully created and stored in SAP HANA.



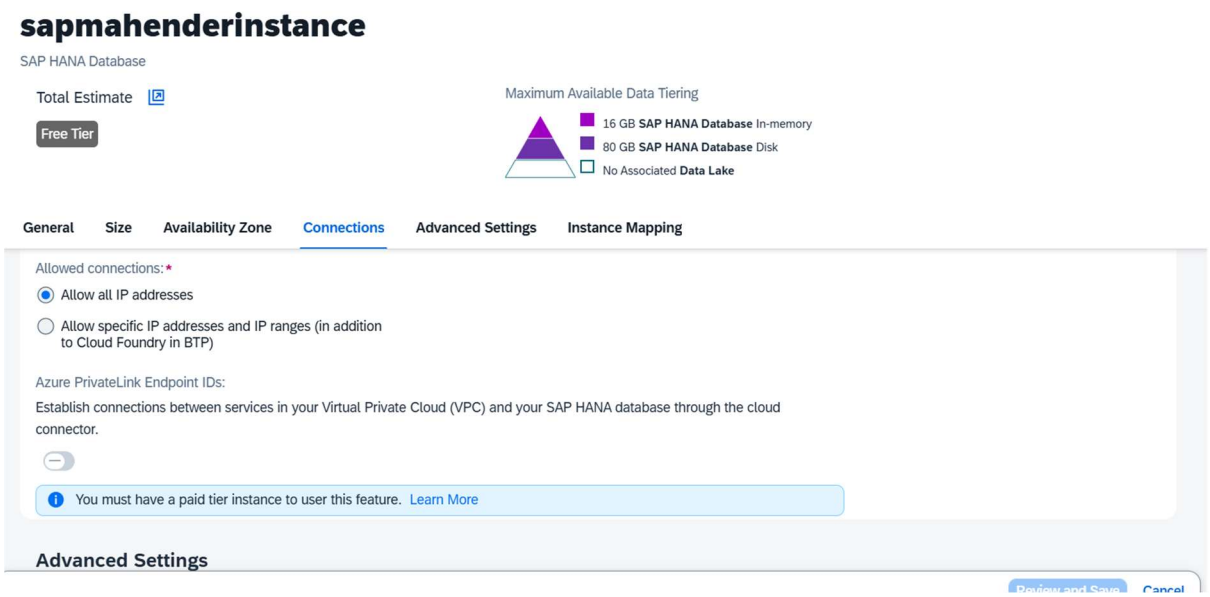
SAP HANA Account Creation



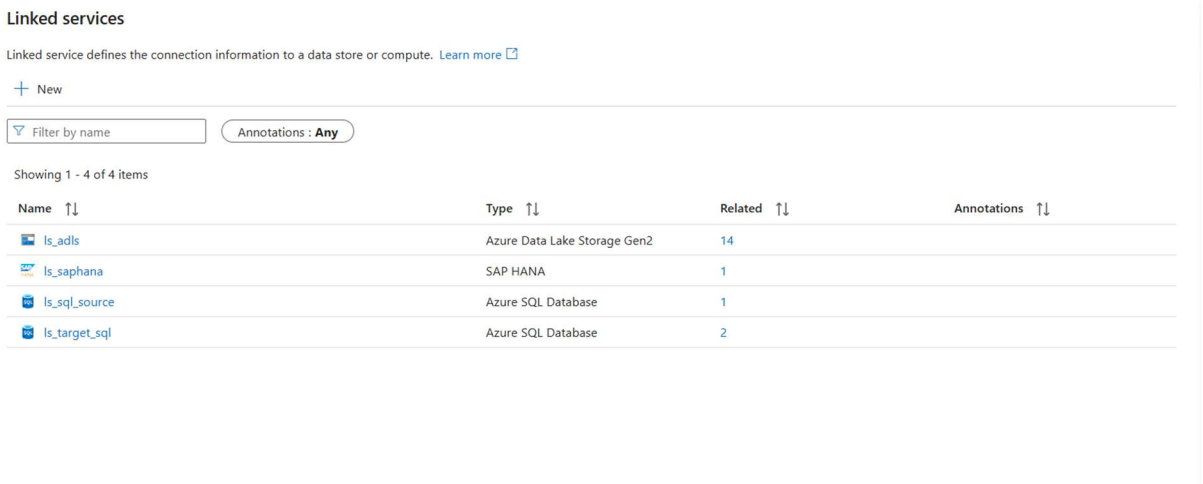
Tables Creation in SAP using Import Wizard

2.2 Connectivity Setup

- Configure Self Hosted Integration Runtime (SHIR)
- Create Linked Service: ls_saphana
- Configure firewall to allow all IP access
- Validate connection from ADF



Connection to allow data transfer in SAP



Linked Services Creation

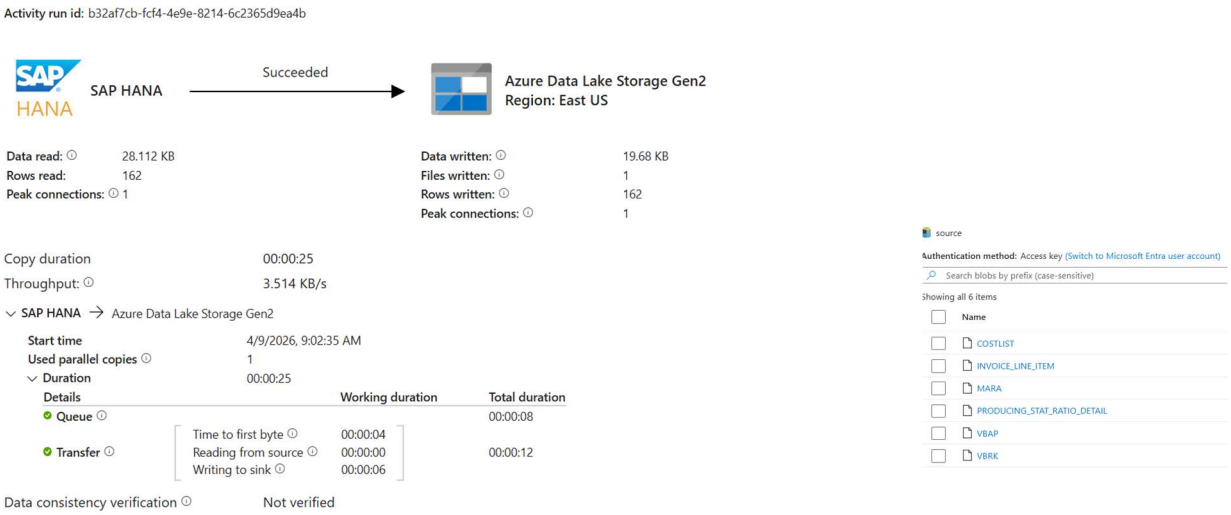
Pipeline 1 – Copy Data from SAP HANA to ADLS Source Container

Extracting data from SAP HANA using Azure Data Factory, where six tables—MARA, VBRK, VBAP, COSTLIST, INVOICE_LINE_ITEM, and PRODUCING_STAT_RATIO_DETAIL—are copied into the source layer of Azure Data Lake Storage (ADLS) using a parameterized pipeline

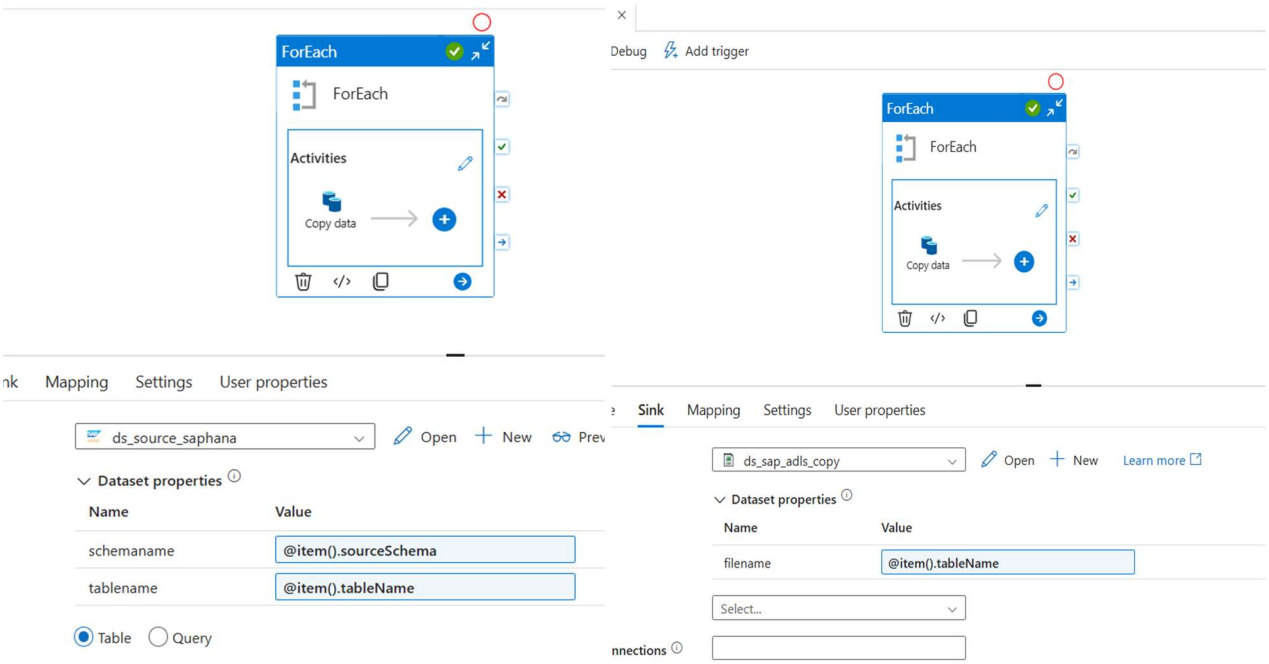
Activities used: Copy Activity, For Each

Parameter used:

[{"tableName":"INVOICE_LINE_ITEM","sourceSchema":"DBADMIN"},{"tableName":"PRODUCING_STAT_RATIO_DETAIL","sourceSchema":"DBADMIN"},{"tableName":"COSTLIST","sourceSchema":"DBADMIN"},{"tableName":"MARA","sourceSchema":"DBADMIN"},{"tableName":"VBRK","sourceSchema":"DBADMIN"},{"tableName":"VBAP","sourceSchema":"DBADMIN"}]



SAP – ADLS – Data Movement



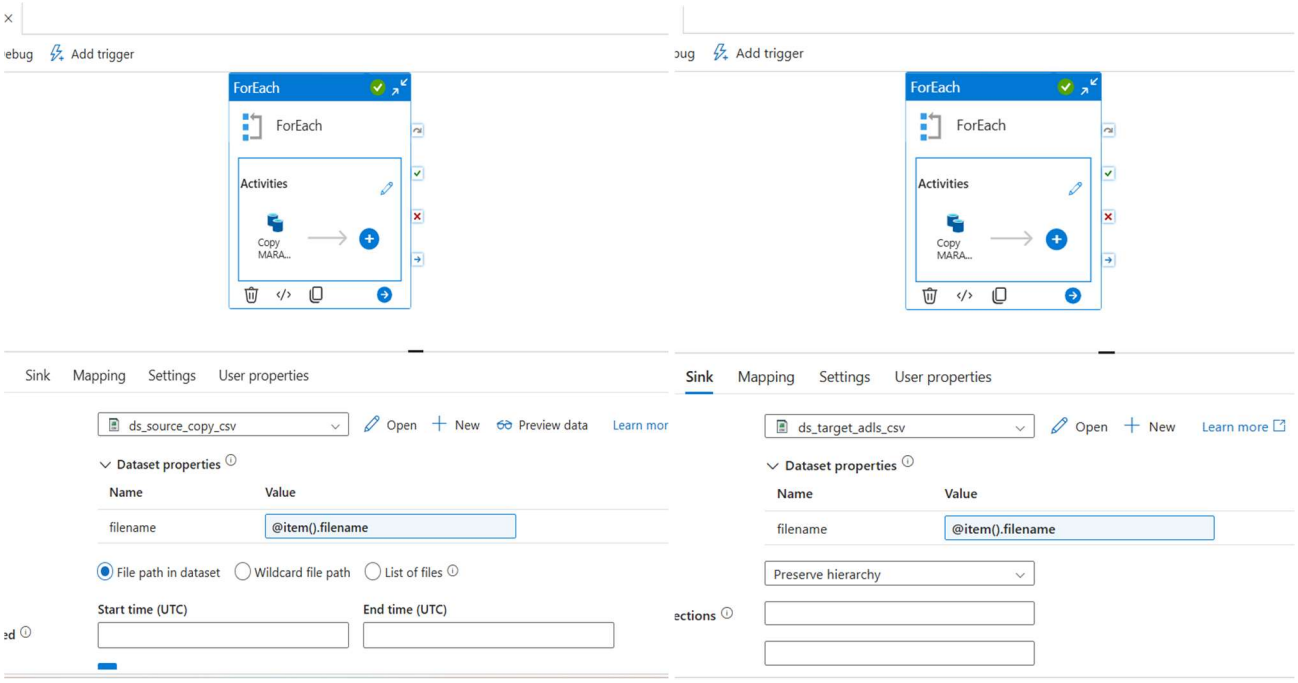
Source and Sink – SAP to ADLS

Pipeline 2 – Copy MARA, VBRK, VBAP to Target Silver ADLS

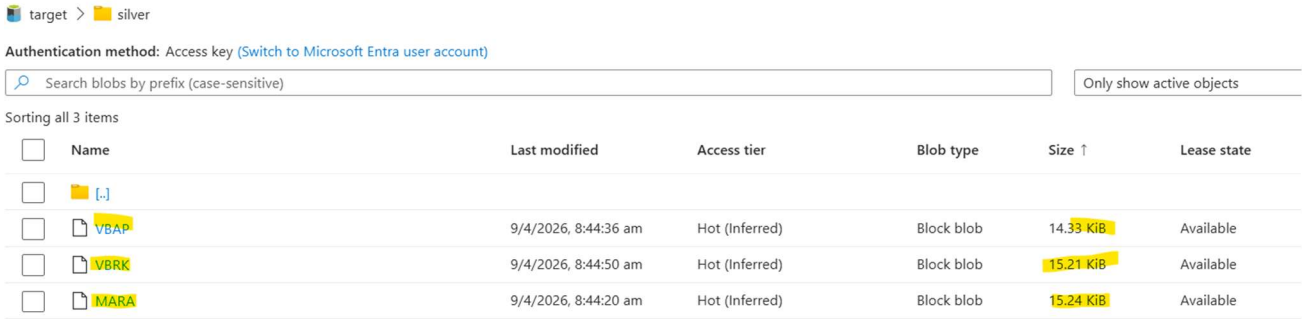
Using the Copy Activity, transferred 3 files from source adls to Target adls (silver container)

Activities used: Copy Activity, For Each

Parameter used: [{"filename":"MARA"},{"filename":"VBAP"},{"filename":"VBRK"}]



Source and Sink – Source to Target ADLS



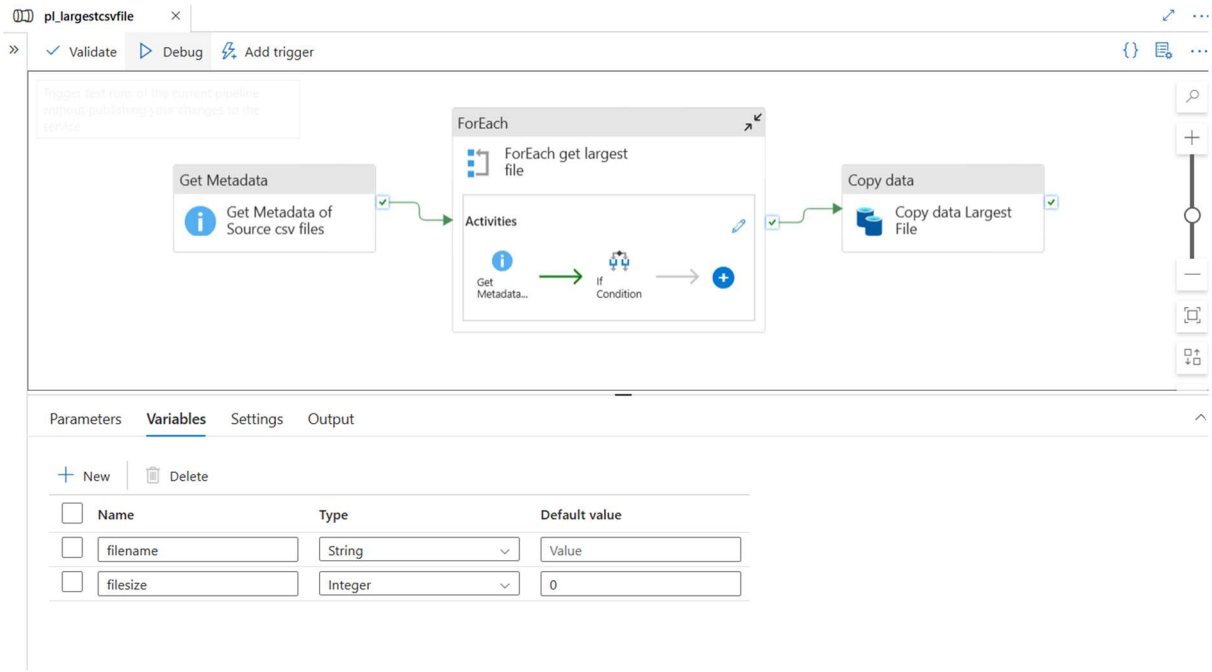
3 Files in Target – silver layer

Pipeline 3 – Copy Largest File in Target (silver) to Gold ADLS

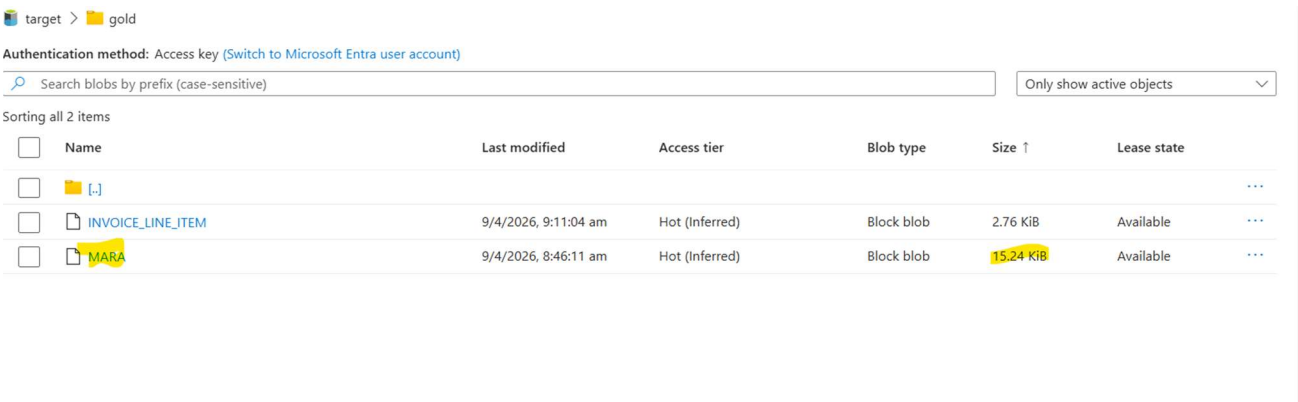
Identifying largest file dynamically and moving to Gold Layer

Activities used: Get Metadata, For Each, If Condition, Set Variables, Copy Activity

Variables used: filename, filesize



Silver – Gold ADLS



Largest File Copied to Gold ADLS

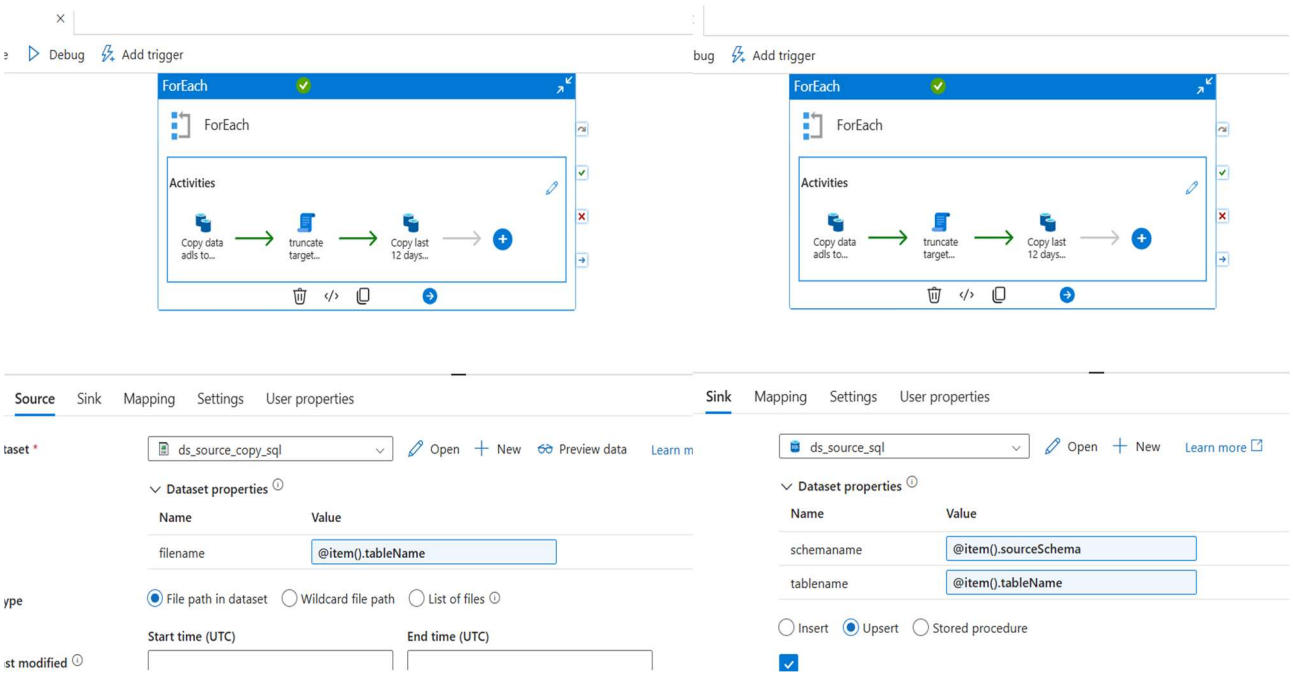
Pipeline 4 – Copy Files to Source SQL & Load Last 12 Days Data to Target SQL

COSTLIST, INVOICE_LINE_ITEM, PRODUCING_STAT_RATIO_DETAIL are ingested from source ADLS into source SQL tables. From there, a transformation step is applied where only the last 12 days of data is filtered using a dynamic SQL condition and loaded into the silver schema in SQL after truncating existing records.

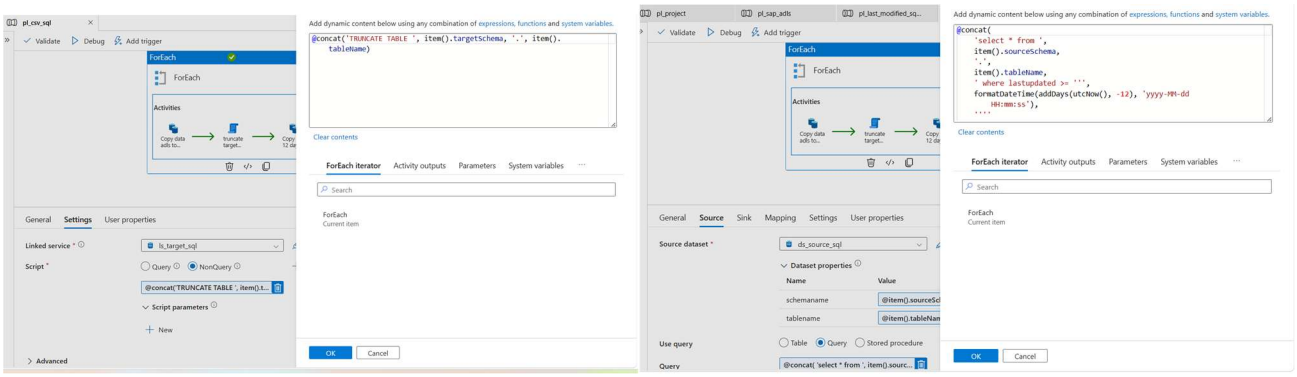
Activities used: For Each, Copy Activity, Script Activity

Parameter used:

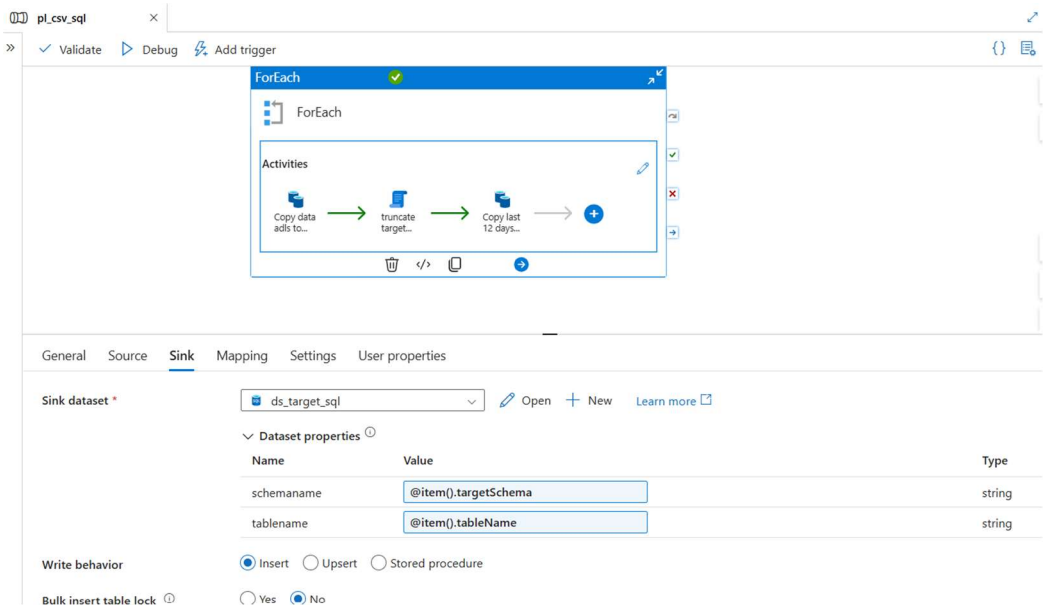
[{"tableName":"INVOICE_LINE_ITEM","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"InvoiceID"},{"tableName":"PRODUCING_STAT_RATIO_DETAIL","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"ProductionID"},{"tableName":"COSTLIST","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"CostID"}]



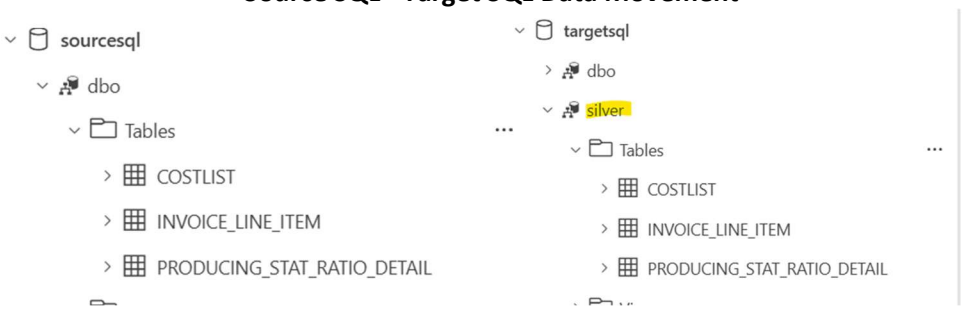
Copy from ADLS to Source SQL



Truncating and ingesting last 12 days data to Target SQL(Silver Schema)



Source SQL - Target SQL Data Movement



Source – Target SQL Tables

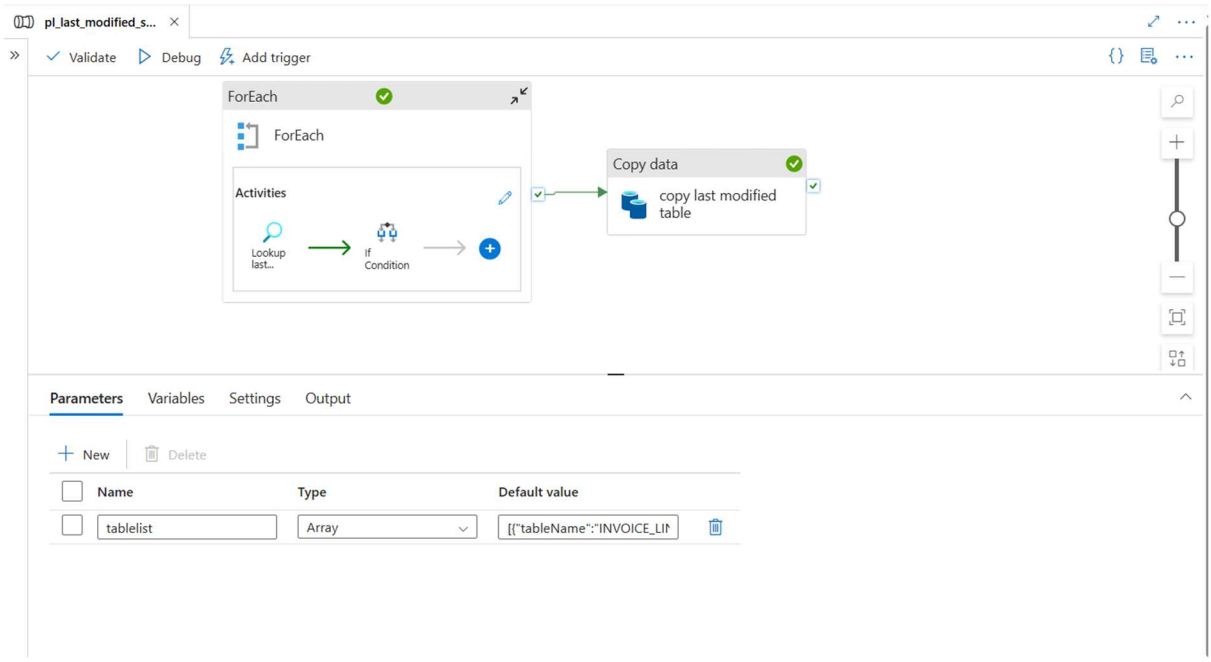
Pipeline 5 – Copy last modified target SQL table to Gold ADLS

Lookup activity is used to identify the latest modified timestamp (MAX LastUpdated) from the Silver tables, and only the most recent table is copied into the Gold layer of ADLS

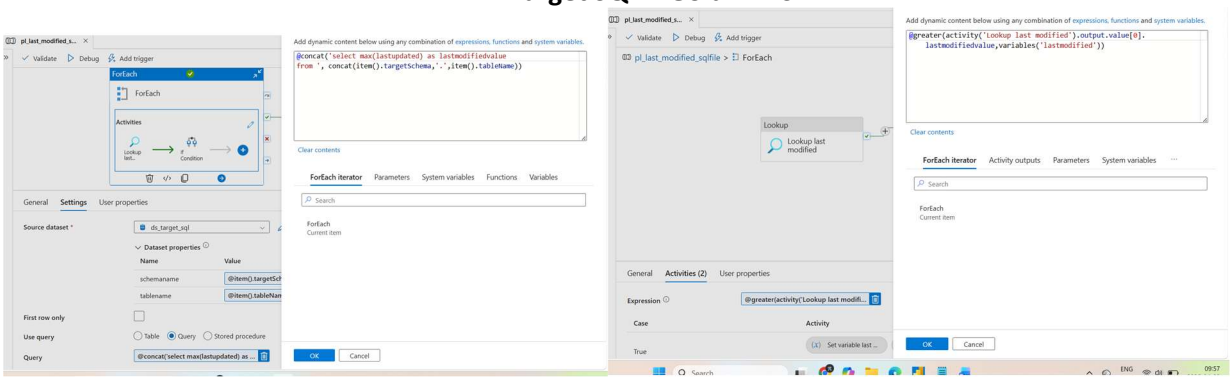
Activities used: For Each, Copy Activity, Lookup, If Condition Activity, Set Variables

Parameter used:
[{"tableName":"INVOICE_LINE_ITEM","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"InvoiceID"},{"tableName":"PRODUCING_STAT_RATIO_DETAIL","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"ProductionID"},{"tableName":"COSTLIST","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"CostID"}]

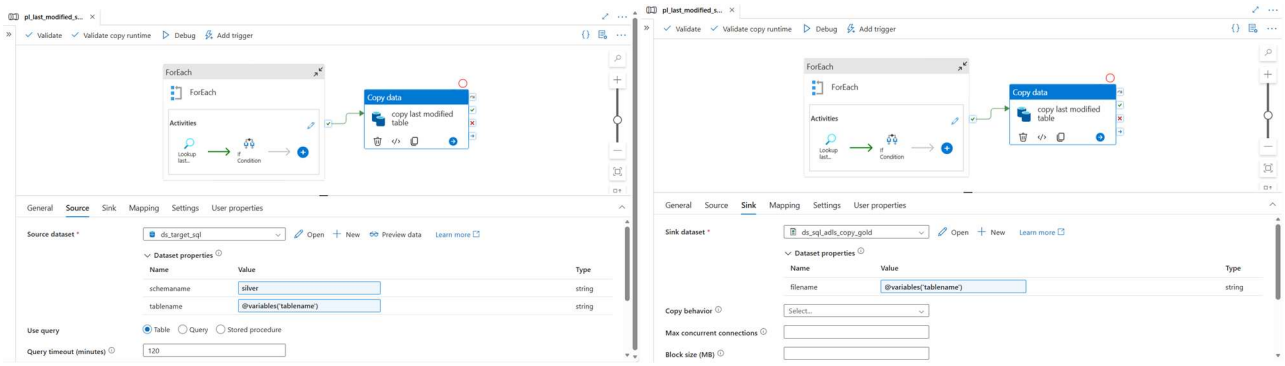
Variables used: tablename, lastmodified



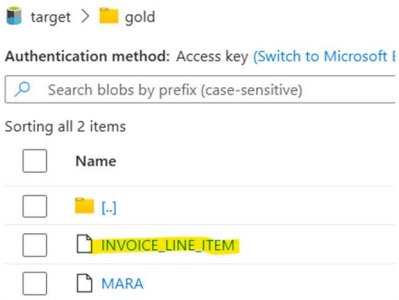
Target SQL – Gold ADLS



Fetching max of Last Updated



Last Modified Table to Gold ADLS



Last Modified Table – Invoice_Line_Item in Gold ADLS

Combined Pipeline – Executing all 5 Pipelines at once with schedule trigger

Parameter used:

tablelist:

[{"tableName":"INVOICE_LINE_ITEM","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"InvoiceID"},{"tableName":"PRODUCING_STAT_RATIO_DETAIL","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"ProductionID"},{"tableName":"COSTLIST","sourceSchema":"dbo","targetSchema":"silver","primaryKey":"CostID"}]

filelist:

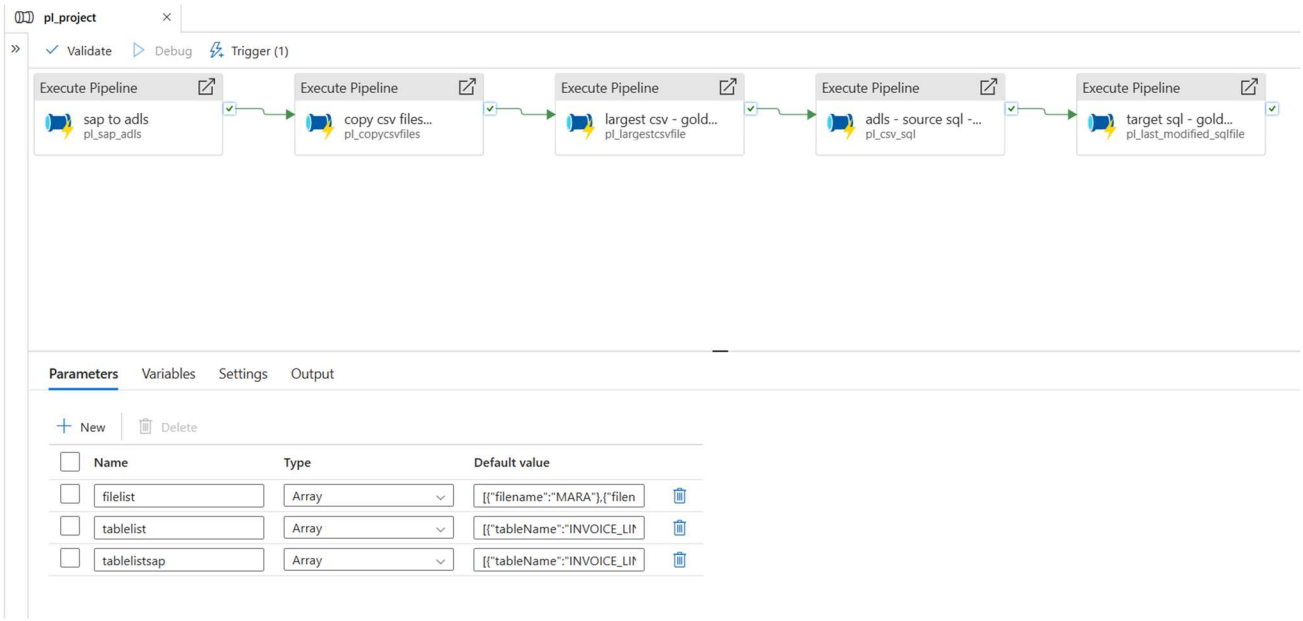
[{"filename":"MARA"},{"filename":"VBAP"},{"filename":"VBRK"}]

tablelistsap:

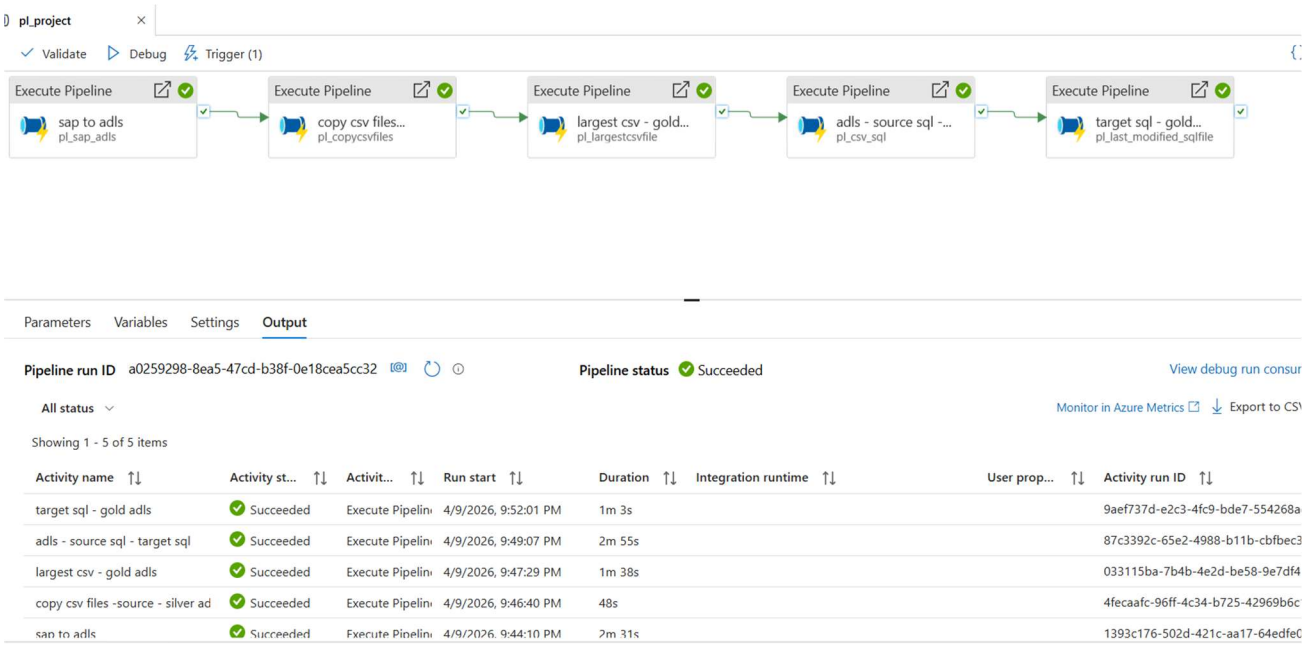
[{"tableName":"INVOICE_LINE_ITEM","sourceSchema":"DBADMIN"},{"tableName":"PRODUCING_STAT_RATIO_DETAIL","sourceSchema":"DBADMIN"},{"tableName":"COSTLIST","sourceSchema":"DBADMIN"},{"tableName":"MARA","sourceSchema":"DBADMIN"},{"tableName":"VBRK","sourceSchema":"DBADMIN"},{"tableName":"VBAP","sourceSchema":"DBADMIN"}]

Trigger used:

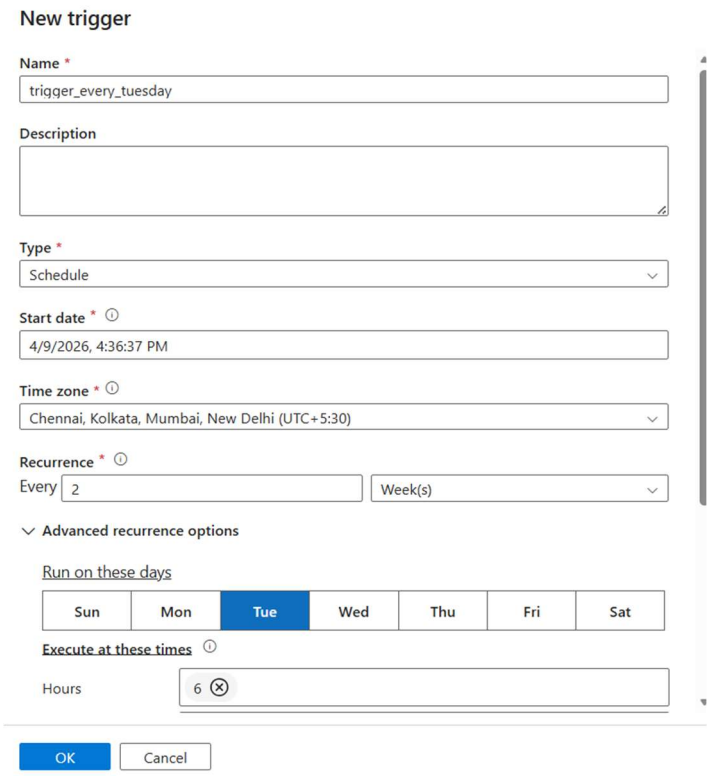
Scheduled trigger every 2nd Tuesday at 6:30



All Five Pipelines



Successfully pipeline run



Trigger to schedule pipeline

Successfully implemented the project

Thank You Sir